PREDICTING JOB CALLBACKS USING RESUME DATA

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INTRODUCTION

- AI Resume Screening = Efficiency in Job Recruitment
- Over **65%** of recruiters have already implemented AI-powered techniques
- Candidate Concerns:
  - Overlooking
  - Introducing bias
  - Allows for objective analysis but can sometimes exist in a **black box** for applicants and recruiters alike

Can ML Models Accurately Depict a Resume Callback Outcome?

01

PROPOSED METHODOLOGY

02

RESULTS AND DISCUSSION

03

CONCLUSION

PROPOSED METHODOLOGY

Our plan to understand why certain resumes succeed



PATTERN DISCOVERY

Identify patterns and resume features that increase fitness and acceptance.



PREDICTIVE MODELING

Apply ML models to classify resumes based on callback probability and analyze accuracy.



VARIABLE ANALYSIS

Analyze the results of the variable testing to determine the accuracy of the ML Models.



PATTERN DISCOVERY

Knowledge Discovery

- **KDD:** Process we use that analyzes datasets to uncover trends
- Help translate raw data → actionable insights



Statistical Testing

- **T-test:** Compare numerical averages between two groups.
- **Chi-square test:** Compare observed frequencies to check for association between variables..
- **Mann-Whitney U (WMW) test:** Compare two independent groups to determine statistical difference.



PREDICTIVE MODELING

Machine Learning Models

- **Predictive modeling:** Use historical data to train algorithms to predict future outcomes.
- **Logistic Regression:** Predict probability of binary outcome
- **Random Forest:** Combines multiple decision trees for single outcome; used in classification and regression.
- **K-Prototype Clustering:** Clustering algorithm for mixed data (numerical and categorical)



Model Evaluation

- Measures how well each model performs to classify and explain outcomes
- **Accuracy:** overall correctness
- **Precision/Recall:** balance false positives and false negatives
- **AUC-ROC:** measure discriminatory quality



VARIABLE ANALYSIS

Principal Component Analysis (PCA)

- **Dimensional reduction** and feature extraction
- Identifies dominant patterns across **Numerical**, **Ordinal**, and **Categorical** features.
- Helps detect redundancy and structure within the resume dataset.

Model Results and Analysis

- Models find relationships between components of language found in datasets.
- Analyze and manipulate language and text to generate measurable data.
- Explore relationships via correlations and statistical testing.
- Assess data quality and logical consistency

IMPLEMENTATION

Data collecting and Cleaning

- Collected synthetic resume dataset (numerical, ordinal, categorical features)
- Removed unrealistic values; ensured consistency between age and experience.
- Encoded variables and prepared 644 clean, usable records.

Exploratory data Analysis

- Ran descriptive statistics and visualizations.
- Conducted t-tests and chi-square tests
- Found years of experience as the only significant numerical difference.

Dimensionality Reduction (PCA)

- Applied PCA to high dimensional encoded data.
- Identified variance spread across many components.
- Determined no single feature dominated the dataset structure.

Predictive Modeling

- Trained Logistic Regression and Random Forest.
- Evaluated accuracy, precision/recall, and ROC-AUC.
- Random Forest performed best (70.5% accuracy, ROC-AUC 0.80

Clustering & Pattern Discovery

- Used K-Prototype to cluster accepted applicants.
- Identified 4 candidate profile groups based on age, experience, education, and skills.
- Clusters revealed meaningful patterns across skill sets and professional backgrounds.

PROCESS

Data Collecting and Cleaning

Exploratory Data Analysis

Dimensionality Reduction

Predictive Modeling

Clustering and pattern discovery



INITIAL DATASET + CLEANING

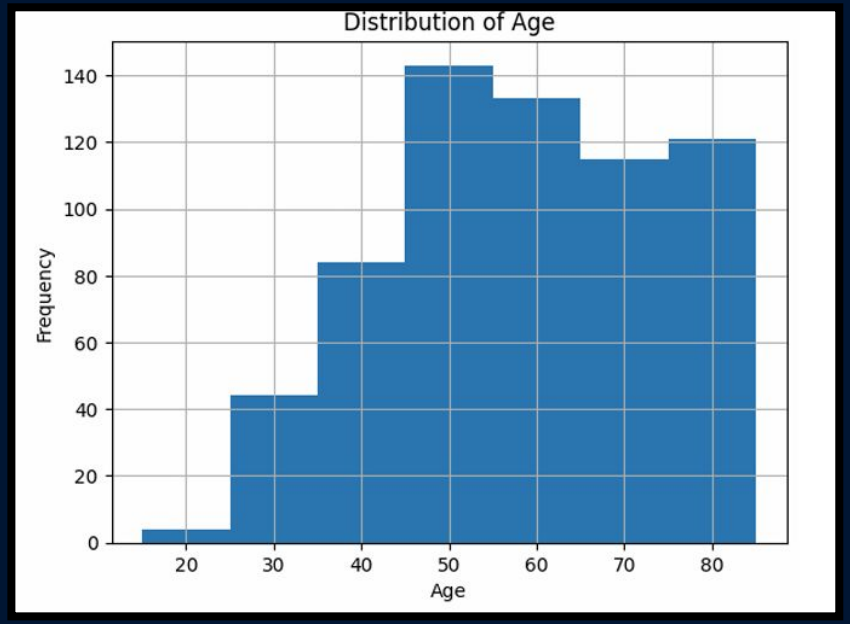
Object/Text-Based Data	Education Level, Skills, Languages Spoken, Technical Proficiencies, Professional Level, Additional Training (First Name, Last Name, Email, Country, Current Job Title, Availability Date, Previous Title Level)
Numerical/Boolean Data	Age, Years of Experience, References Available, GitHub Contributions, Job Accepted

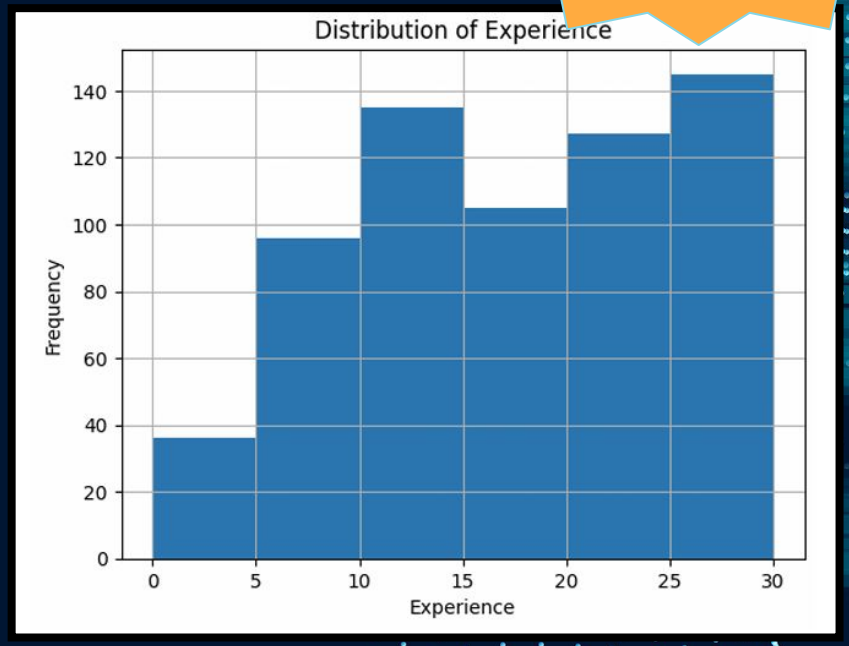
- ✅ Age <= experience
- ✅ Work legally (after age 16)
- ✅ + Age and + Experience
- ✅ If > 400 GH, 2+ experience OR Master's/PhD
- ✅ If > 700 GH, 4+ experience OR Master's/PhD
- ✅ Meet age requirements by professional level
- ✅ Meet experience requirement by professional level
- ✅ Meet age requirement by educational level
- ✅ Consistent format, no duplicates, no null values

644 valid entries

EDA

Not normally distributed

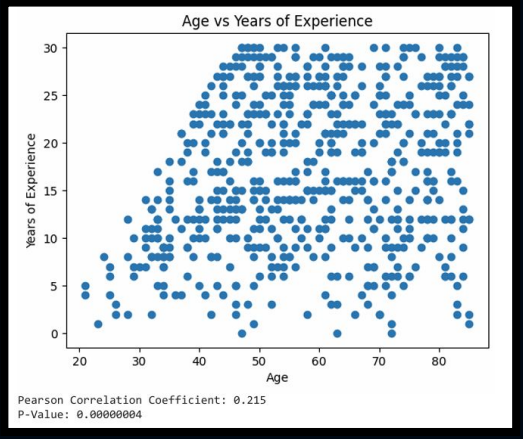




Least = youngest (15-25)
Most = 44-55, 55-65 (older, overrepresented)

Least = novice (0-5)
Most = experienced (25-30)
Longer careers are overrepresented

CORRELATION TESTS

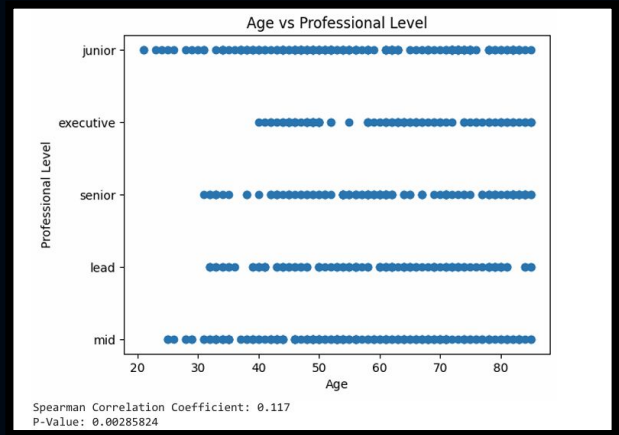


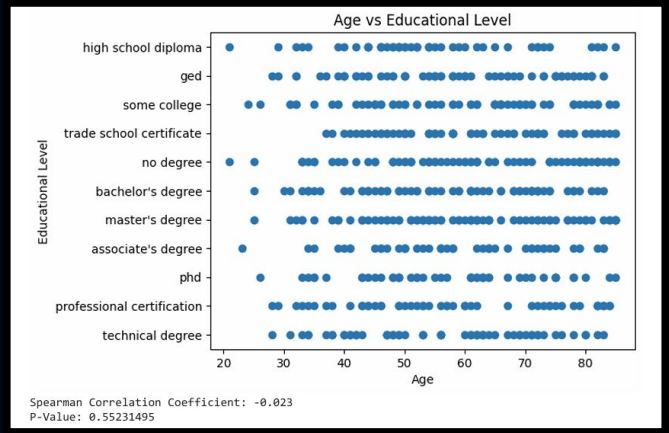
Poor clustering/trends

Weak, statistically significant correlation: as age increases, years of experience increase

Fairly uniform distribution

Weak, statistically significant correlation: as age increases, professional level increases; negligible





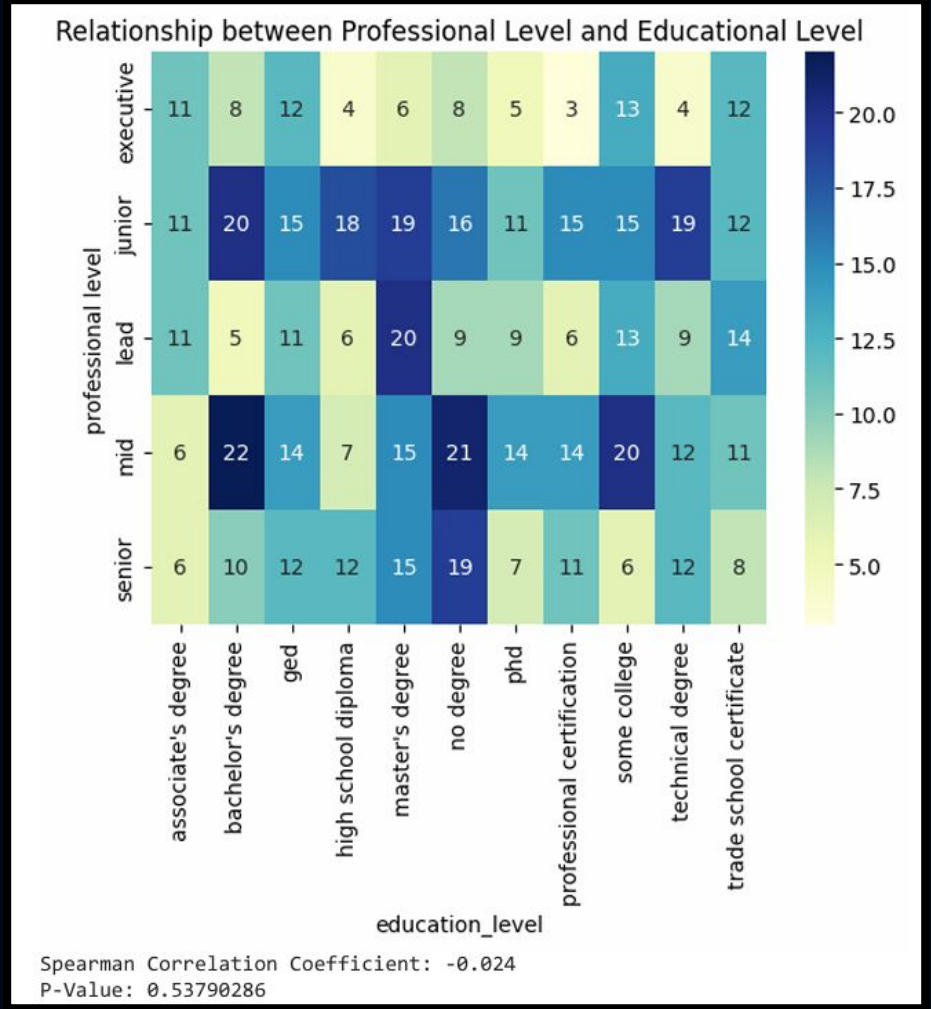
Fairly uniform distribution

Weak, statistically insignificant negative correlation between age and educational level

CORRELATION TESTS

Fairly random distributions

Weak, statistically insignificant relationship between professional level and educational level



SKILLS BY PROFESSIONAL LEVEL

	Junior	Mid	Senior	Lead	Executive
Skills	Graphic Design, Time Management, Communication	Foreign Languages, Public Speaking, Data Analysis	Writing, Marketing, Event Planning	Financial Analysis, Customer Service, Data Analysis	Networking, Project Management, Event Planning
Technical	CSS, AWS, React	AWS, HTML, Ruby	Jenkins, Vue.js, Git	Git, CSS, TypeScript	Python, Ruby, TensorFlow
Additional Training	Networking, Technology Training, Health and Wellness	Health and Wellness, Time Management, Emotional Intelligence	Emotional Intelligence, Networking, Decision Making	Time Management, Creative Thinking, Cultural Sensitivity	Time Management, Technology Training, Emotional Intelligence

❓ Sparse data - no conclusions re: skills, technical proficiencies

✅ Additional training distributions differ by professional level

RESUME ACCEPTANCE

413 accepted vs **231** rejected resumes

Statistically significant difference in distribution:

- Years of experience

No statistically significant results:

- Age of applicant
- Number of GitHub contributions
- Educational level (⇧ trade school 75.44%, ⇩ no degree 23.39% callback rate)
- References available (⇧ available 65.11%, ⇩ not available 63.12% callback rate)
- Professional level (⇧ executive and lead 100%, ⇩ mid 42.3% callback rate)
- Skills, technical proficiencies, or additional training
- Languages spoken

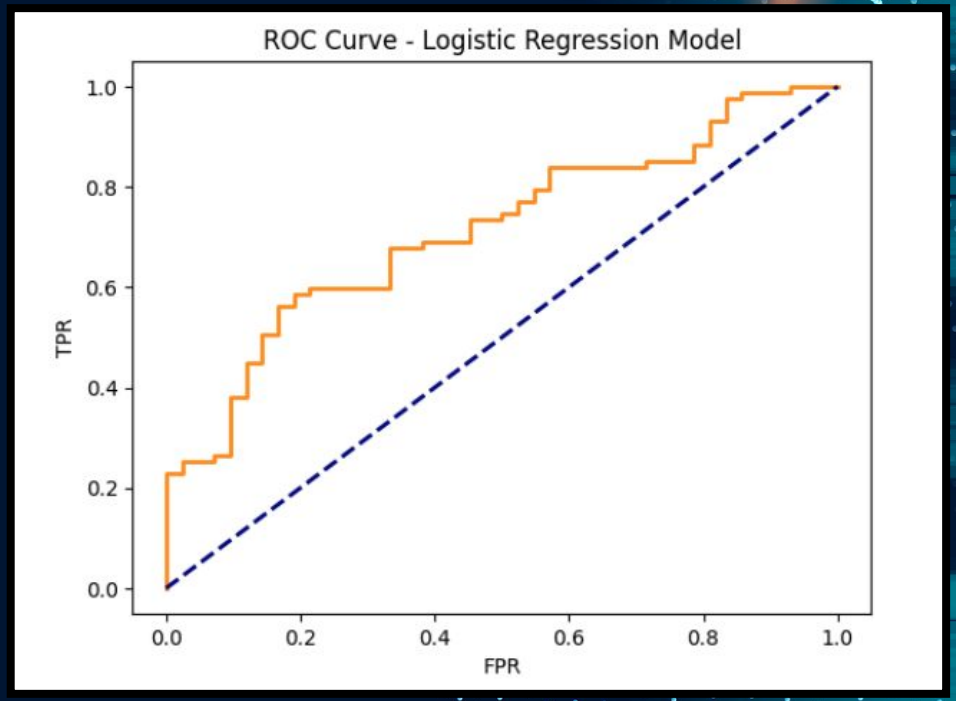
PRINCIPAL COMPONENT ANALYSIS

Principal Components #1 through #4 explain appx. **48.9%** of the total variance in the dataset



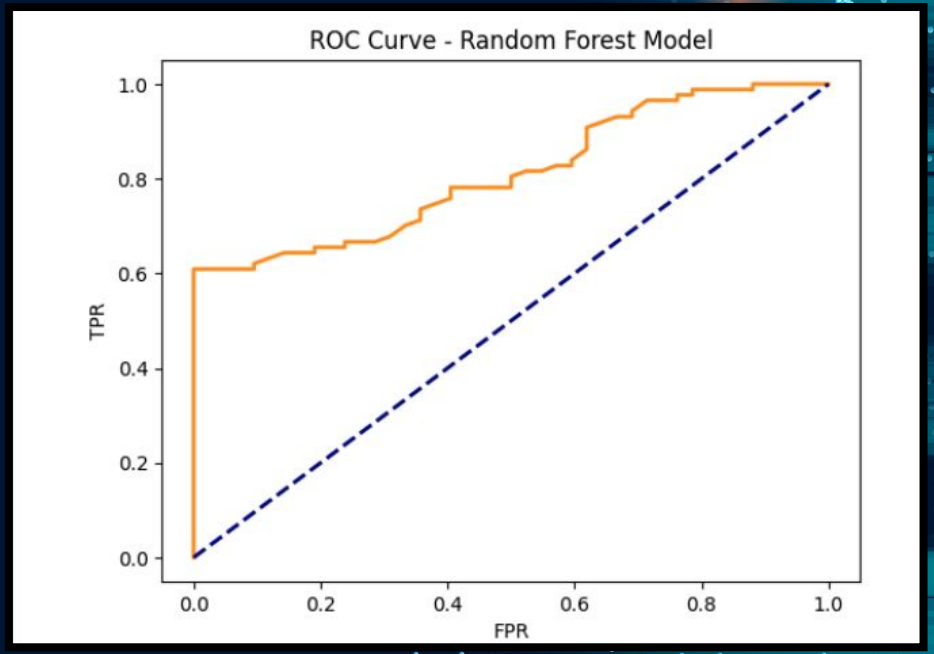
LOGISTIC REGRESSION

- Preprocess w/ ColumnTransformer (StandardScaler → numerical, OHE → categorical)
  - 66.67% accurate
  - ROC Score 0.7
  - Misclassification of both FN and FP from Confusion Matrix



RANDOM FOREST MODEL

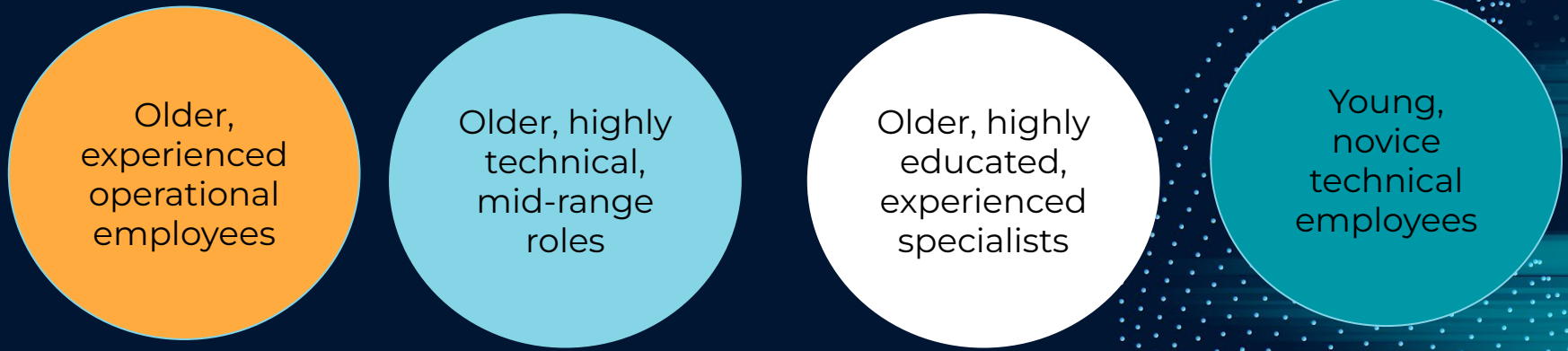
- Preprocess w/ ColumnTransformer (StandardScaler → numerical, OHE → categorical)
  - 70.54% accurate
  - ROC Score 0.81
  - Acceptable model with predictive power
  - Outperforms logistic regression



K-PROTOTYPES CLUSTERING

Data Splitting → Mask to only accepted resumes → Preprocess → Recombine data

Determine optimal number of clusters (Elbow Method) → Fit model → Deploy → Interpret clusters



CONCLUSION

This project analyzed the relationship between resume attributes and callback outcomes using statistical analysis, machine learning, PCA, and clustering.

- Experience was the only statistically significant standalone predictor; age and GitHub contributions were not meaningful.
- Categorical variables had limited individual impact due to the synthetic dataset.
- Random Forest outperformed Logistic Regression with higher accuracy and stronger ROC-AUC by capturing nonlinear interactions.
- PCA showed variance is spread across many resume features, indicating a multidimensional structure.
- K-Prototype clustering identified four distinct candidate profiles based on combinations of age, experience, education, skills, and training.
- No single factor determines acceptance; callback patterns emerge from combined attributes.

This demonstrates that resume evaluation is multifactorial, requiring a combination of statistical testing, dimensionality reduction, predictive modeling, and clustering. Despite the synthetic data, the workflow offers a strong framework for real-world resume analysis and emphasizes interpretability, fairness, and multidimensional analysis in AI hiring systems.

THANKS!

Do you have any questions?

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